



Unbiased estimator of Bernoulli distribution

Asked 5 years, 6 months ago Modified 5 years, 6 months ago Viewed 6k times



1



Let $X_1, \dots, X_n$ be a random sample from Bernoulli (p), where $p \in (0,1)$ is unknown.

(a) Find an unbiased estimator W of $\tau(p) = p(1 - p)$.

(b) Find an unbiased estimator which has smaller MSE than W .

My approach:

I know sample variance is unbiased estimator of population variance. Hence $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of $p(1 - p)$. But I don't know how to find an unbiased estimator of W . Can someone assist me in answering (a) and (b)?

statistics

statistical-inference

Share Cite Edit Follow Flag

asked Mar 31, 2019 at 15:39



Lady

401

7

16



it seem, X_i are independent. – Masoud Mar 31, 2019 at 18:29



Yes... they are independent. – Lady Mar 31, 2019 at 18:48



Related: math.stackexchange.com/q/3086919/321264. – StubbornAtom Apr 1, 2019 at 13:46



1 Answer

Sorted by: Highest score (default)



1



(a)

$$W = X_1(1 - X_2)$$

$$E(W) = E(X_1(1 - X_2)) = E(X_1)E(1 - X_2) = p(1 - p)$$

(b)

$T = \sum_{i=1}^n X_i$ is complete and sufficient estimator so $E(X_1(1 - X_2)|T)$ is UMVUE for $p(1 - p)$

X_1	X_2	$X_1(1 - X_2)$
0	0	0
0	1	0
1	0	1
1	1	0

so $X_1(1 - X_2)$ is a Bernoulli random.

$$p(X_1(1 - X_2) = 1|T = t) = p(X_1 = 1, X_2 = 0 | \sum_i X_i = t)$$

$$= \frac{P(X_1=1, X_2=0, \sum_{i=3}^n X_i=t)}{P(\sum_{i=1}^n X_i=t)}$$

$$= \frac{P(X_1=1, X_2=0, \sum_{i=3}^n X_i=t-1)}{P(\sum_{i=1}^n X_i=t)}$$

$$= \frac{pqP(\sum_{i=3}^n X_i=t-1)}{P(\sum_{i=1}^n X_i=t)}$$

$$= \frac{pq \binom{n-2}{t-1} p^{t-1} q^{n-2-t+1}}{\binom{n}{t} p^t q^{n-t}} =$$

$$= \frac{\binom{n-2}{t-1}}{\binom{n}{t}} \text{ (for } t \geq 1 \text{ O.W equal zero)}$$

$$E(X_1(1 - X_2)|T = t) = 0 + 1 \times \frac{\binom{n-2}{t-1}}{\binom{n}{t}} = \frac{\binom{n-2}{t-1}}{\binom{n}{t}} = \frac{t(n-t)}{n(n-1)}$$

$$\text{so UMVUE for } p(1-p) = \frac{\sum X_i(n - \sum X_i)}{n(n-1)} = \frac{n\bar{X}(n - n\bar{X})}{n(n-1)} = \frac{n\bar{X}(1 - \bar{X})}{(n-1)}$$

$$\text{also it equal to } \frac{1}{n-1} \sum (X_i - \bar{X})^2$$

$$\frac{1}{n-1} \sum (X_i - \bar{X})^2 = \frac{1}{n-1} (\sum X_i^2 - n\bar{X}^2)$$

since $X_i \in \{0, 1\}$

$$= \frac{1}{n-1} (\sum X_i - n\bar{X}^2)$$

$$= \frac{1}{n-1} (n\bar{X} - n\bar{X}^2) = \frac{n}{n-1} \bar{X}(1 - \bar{X})$$

at this step since $S^2 = \frac{n}{n-1} \bar{X}(1 - \bar{X})$ so S^2 is a function of $\bar{X}$ (sufficient and complete estimator) and also is unbiased so directly it is UMVUE for $p(1-p)$ (and this is another proof for part(b))

Share Cite Edit Follow Flag

edited Apr 1, 2019 at 10:54

answered Mar 31, 2019 at 16:37



Masoud

2,745

2

8

23



is it clear or need explanation? – Masoud Mar 31, 2019 at 18:49

